

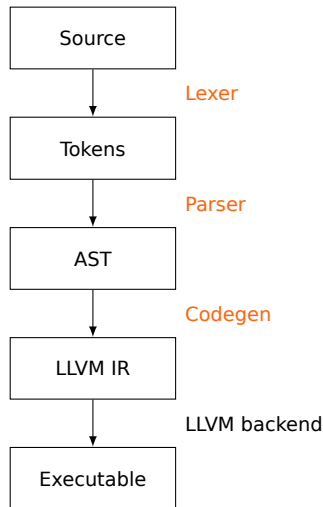
Compiler implementation

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What is this course about?

- Building **your own compiler** step by step
 - Use **any language** you like
 - Implement **Lexer**
 - Implement **Parser**
 - Implement **LLVM IR generation**
 - Use LLVM framework as backend
- Incremental evolution of source language
 1. Basic untyped expression language
 2. Control flow constructs
 3. Functions
 4. Primitive type system
 5. Structs and arrays
- Theory and practice together
More compiler theory on elective courses!
- Next semester will re-use developed compiler



Structure

- ~14 weeks, 1.5 classes per week
- Mixed lecture-seminar format
- 5 language extensions to implement
 1. Basic untyped expression language (6 weeks)
 2. Control flow constructs (2 weeks)
 3. Functions (2 weeks)
 4. Primitive type system (2 weeks)
 5. Structs and arrays (2 weeks)
- Final grade is based on implemented extensions
 - **5** — all 5 extensions
 - **4** — 4 out of 5 extensions
 - **3** — 3 out of 5 extensions



<https://nsu-syspro.github.io/courses/translators/>

Q&A